

The Gravitational Register of Sun, Earth, Jupiter, Kepler-452b

A personal monograph of the PLANETVORTEX closed gravifigure

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Table of Contents

1. The model in one page

The Klein quartic $x^3y+y^3z+z^3x=0$ is the genus-3 surface whose automorphism group $PSL(2,7)$ of order 168 attains the Hurwitz bound $84(g-1)$. Its $\{7,3\}$ tessellation carries 24 heptagons — 12 antipodal pairs — and stage V2 of the bench carries the mass ladder of the chosen bodies in the vertex angles: $\alpha_i=2\pi/3-\sigma s_i$ with $s_i=\ln GM_i-\langle \ln GM \rangle$. The geometric-mean normalization makes $\sum s_i=0$ exactly, and that is precisely the condition that the decorated tessellation closes on the Gauss–Bonnet budget $\sum \alpha_i=8\pi=2\pi(2g-2)$: the Euler characteristic absorbs the mass ladder. The heavier the body, the wider its heptagon — gravity drawn as geometry, in exact closed form: $\cosh R_i=\cot(\alpha_i/2)\cot(\pi/7)$, $\cosh(\ell_i/2)=\cos(\pi/7)/\sin(\alpha_i/2)$, $\cosh r_i=\cos(\alpha_i/2)/\sin(\pi/7)$, $A_i=5\pi-7\alpha_i$.

2. The gravimetric register algebra of the chosen set

2.1
The
exact
registe
r
dimen
sions

body
GM, m³/s²
s

α , rad

R

ℓ

r

A

i, °

λ , rad

Sun

1.327124e+20

+8.2455

1.863101

1.000465

0.980655

0.841019

2.666254

—

—

Earth

3.986004e+14

-4.4703

2.219790

0.248924

0.217815

0.223396

0.169432

0.0000

0.000000

Jupiter

1.266865e+17

+1.2913

2.058174

0.692558

0.639246

0.604496

1.300744

1.3044

0.022766

Kepler-452b

2.195845e+14

-5.0665
2.236515
0.138342
0.120360
0.124492
0.052360

—
—

A custom body: $GM = 2.195845e+14 \text{ m}^3/\text{s}^2$ — registered locally for this monograph only.,
radius 1.63 km

$\lambda = \bar{R} \cdot i$ is the arc register of the spatial tilt against the ecliptic ($\bar{R} = 1 \text{ AU}$); the inclination
rows come from the committed J2000 register (JPL).

2.2 The budget closure of this set

The budget closure of this set: $\Sigma\alpha = 8.377580409573$ against $8\pi = 25.132741228718$ —
residual $1.68e+01$ (**the figure does NOT close (check the normalization)**).

3. The spatial registers (the inclined sky)

The spatial registers of the bench: the tilt operators $T = R_z(\Omega) R_x(i) \in SO(3)$, the arc
registers $\lambda_i = \dot{R} i_i$ and the mutual-inclination matrix are certified by stage V1 (protocol
V1_inclined_registers.json); the 3D Newtonian run from the real J2000 sky carries the
conservation of the energy and of the full angular-momentum vector.

4. The

verification certificate of this run

This monograph was generated together with a live verification run; the checks below were executed by the generator, not quoted.

#

check

result

recorded

1

$\Sigma\alpha = 8\pi$ (mpmath, 30 dps)

PASS

0.0

2

$\Sigma\alpha = 8\pi$ (float64)

PASS

7.11e-15

3

the hyperbolic Pythagoras (12 cells)

PASS

3.74e-16

4

the $SO(3)$ tilt orthogonality

PASS

2.22e-16

5

protocol P4_fano_algebra

PASS

committed

6

protocol X4_hyperbolic_figure

PASS

committed

7

protocol V2_klein_tiling

PASS

committed

5. The corpus and the reproduction commands

The formal corpus: Theorems A–C, Lemma D, Lemma E (the planar bench), Theorem F (the coset construction), Lemma F (the antipodal freeness), Theorem G (the budget closure), Lemma G (the arc register) — docs/theorems/.

Every plotted or quoted number of the bench is bound to a JSON protocol in results/protocols/ — committed before the run’s tolerance, refreshed after it.

Reproduce everything with the committed pipeline (from the mini-repository root):

`make all-ladders` *# P1..P7 + X1..X6 + V1..V2, JSON protocols*

`make test` *# the pytest guard*

`make crosslang` *# the C99 kernel vs Python*

`make figures600` *# this gallery at 600 dpi*

`make monograph` *# a sample monograph*